Journal of Breast Imaging, 2026, Volume 8, Number 4, 354–365
doi:10.1093/jbi/wbag025
Science of Screening
Received: February 4, 2026; Accepted: March 27, 2026
Published Online: July 21, 2026

Science of Screening

Contrast-Enhanced Mammography for Breast Cancer Screening: A Systematic Review and Meta-Analysis

Komal Verma Saluja, MBBS,[1,*] Sangeeta Saxena, MD,[1] Harshvardhan Khokhar, MD,[1] Ashok Kumar Sharma, MD,[1] Dharmraj Meena, MD,[1] Atul Sharma, MS,[3] Mayank Goyal, MD[2]

[1]Department of Radiology, Government Medical College, RUHS, Kota, Rajasthan, India, [2]Department of Radiology, All India Institute of Medical Sciences, New Delhi, India, [3]Department of Surgery, Government Medical College, RUHS, Kota, Rajasthan, India

[*]Corresponding author: Komal Verma Saluja (komalvermasaluja@gmail.com)

Abstract

Objective: To evaluate pathway-specific cancer detection and summarise screening outcomes (recall/biopsy/PPV3) for contrast-enhanced mammography when reported, and to compare them with those for low-energy mammography and breast MRI.

Methods: This systematic review and meta-analysis followed PRISMA guidelines. PubMed (MEDLINE), Embase, Scopus, and Web of Science were searched from January 1, 2011, through August 31, 2025, for studies evaluating contrast-enhanced mammography in screening or surveillance populations (including supplemental/add-on screening and surveillance of patients with a personal history of breast cancer). Diagnostic, recall, and problem-solving studies were excluded. Screening pathways were categorized as: (1) direct comparison with low-energy mammography, (2) direct comparison with MRI, and (3) add-on CEM to mammography-based screening. Random-effects meta-analyses were performed when designs were comparable. Comparisons between CEM and MRI were summarized qualitatively because of heterogeneity and limited cancer events across studies. Single-arm screening studies were summarized descriptively. Risk of bias was assessed using Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2).

Results: Fourteen studies met eligibility criteria for qualitative synthesis; seven screening and surveillance studies contributed to the meta-analysis. Compared with low-energy mammography, CEM detected 9.3 additional cancers per 1,000 women screened (risk difference [RD], +9.3 per 1,000; 95% CI, 4.0–14.6; $P < 0.01$; $I^2 = 0\%$). Three studies directly compared CEM with MRI in screening settings and were summarized qualitatively because of heterogeneity and limited cancer events.

Conclusion: In screening and surveillance populations, contrast-enhanced mammography increases cancer detection compared with low-energy mammography. Evidence from available comparative studies suggests that cancer detection with CEM may be broadly comparable to MRI in selected screening contexts; however, the limited number of studies and heterogeneous designs preclude definitive comparative conclusions. These pathway-specific estimates may inform clinical implementation and the design of prospective screening trials.

Key words: contrast-enhanced mammography; breast cancer screening; supplemental screening; magnetic resonance imaging; systematic review.

© The Author(s) 2026. Published by Oxford University Press on behalf of Society of Breast Imaging.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (https://creativecommons.org/licenses/by/4.0/), which permits unrestricted reuse, distribution, and reproduction in any medium, provided the original work is properly cited.

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026

Key Messages

- In screening and surveillance populations, contrast-enhanced mammography (CEM) demonstrated higher cancer detection than low-energy mammography, with consistent findings across sensitivity analyses.
- A small number of studies have directly compared CEM with breast MRI in screening settings. Across these studies, cancer detection appeared broadly comparable, although quantitative synthesis was not performed because of limited cancer events and heterogeneity.
- Increased cancer detection with CEM was accompanied by higher recall and biopsy rates, emphasizing the importance of balancing potential screening benefits against downstream harms.

Introduction

Breast cancer is the most commonly diagnosed malignancy among women worldwide, with approximately 2.3 million new cases reported in 2022.[1] In the United States, an estimated 316,950 new invasive breast cancers were projected in 2025, underscoring the continued importance of effective population-based screening.[2] Mammography remains the cornerstone of screening and has contributed to reduced breast cancer mortality through earlier detection.[3] However, mammographic performance is reduced in women with dense breast tissue.[4] Dense tissue influences screening outcomes through two distinct mechanisms: a masking effect, in which fibroglandular tissue obscures lesions and reduces mammographic sensitivity, and an independent risk effect, whereby women with dense breasts have a higher underlying incidence of breast cancer and therefore higher interval cancer rates.[5] In recognition of these limitations, the U.S. Food and Drug Administration now mandate standardized breast density notification with mammography to inform patients and clinicians about the potential need for supplemental imaging.[6]

Contemporary screening guidelines from the American Cancer Society[7] and the U.S. Preventive Services Task Force[8] recommend mammography, including digital breast tomosynthesis (DBT), as the foundation of screening for average-risk women, with supplemental imaging primarily reserved for those at elevated risk. The American College of Radiology recommends annual screening beginning at age 40 for average-risk women and annual breast MRI for women at higher-than-average risk, including those with genetic predisposition, prior chest irradiation, or a calculated lifetime risk of 20% or greater.[9] Women with a personal history of breast cancer (PHBC), particularly those with dense breasts, are also commonly considered for supplemental MRI; when MRI is not feasible, contrast-enhanced mammography (CEM) may be considered as an alternative supplemental modality.[10] Collectively, these recommendations highlight the central role of risk stratification and breast density in modality selection for breast cancer screening.

Breast MRI is the most sensitive contrast-based modality for breast cancer screening and is recommended for women at elevated risk.[11,12] However, its use in routine screening is limited by cost, scanner availability, longer acquisition times, contraindications to MRI or gadolinium-based contrast agents, and variable patient tolerance.[13] These constraints have increased interest in more accessible contrast-based imaging alternatives. Contrast-enhanced mammography combines low-energy mammography with intravenous iodinated contrast to provide functional information analogous to MRI, potentially improving lesion conspicuity in dense breast tissue.[14] Emerging screening studies suggest that CEM increases cancer detection compared with mammography alone and may provide similar cancer detection to MRI in selected cohorts, while offering shorter examination times than standard MRI protocols and potentially broader implementation in some practice settings than MRI.[15] CEM involves a higher radiation dose than standard mammography and requires iodinated contrast; reported doses remain within regulatory limits, and serious contrast-related adverse events are uncommon.[16]

A key limitation of the CEM evidence base is that many studies have been performed in diagnostic or problem-solving settings, including symptomatic populations or women recalled after abnormal screening findings. Such cohorts have a higher disease prevalence than screening populations, which can inflate estimates of test performance and limit the generalizability of results to screening programs.[15] This distinction is crucial for guideline and policy decisions, where benefits in cancer detection must be weighed against recall, biopsy, radiation exposure, and resource utilization at the population level.[17,18]

Despite increasing clinical adoption of CEM as a supplemental imaging tool, its role in population-based screening remains uncertain. Prior systematic reviews and meta-analyses have frequently pooled heterogeneous indications (diagnostic evaluation, problem-solving, and preoperative staging) and combined fundamentally different screening pathways, thereby limiting the interpretability of pooled estimates for screening practice.[15]

To address these gaps, we conducted a systematic review and meta-analysis restricted to screening and surveillance studies evaluating CEM. We evaluated three clinically relevant pathways: (1) direct comparison with low-energy mammography, (2) direct comparison with breast MRI, and (3) add-on CEM to mammography-based screening. Single-arm screening studies were summarized descriptively to contextualize absolute cancer detection without inappropriate quantitative pooling.

Methods

Study design and reporting standards

This systematic review was developed from the protocol registered on PROSPERO (CRD420251034142). It was

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026

conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[19] The objective was to evaluate CEM in screening and surveillance contexts, with quantitative synthesis restricted to comparable screening designs.

Literature search strategy

A comprehensive search of PubMed (MEDLINE), Embase, Scopus, and Web of Science was conducted for studies published between January 1, 2011, and August 31, 2025. Search terms included controlled vocabulary and keywords related to CEM and screening (eg, contrast-enhanced mammography, contrast-enhanced spectral mammography, dual-energy mammography, screening, supplemental screening, surveillance, digital mammography, DBT, and magnetic resonance imaging, as well as cancer detection rate and incremental cancer detection rate). Reference lists of eligible studies and relevant reviews were screened for additional studies.

Eligibility criteria

Studies were included if they evaluated CEM in asymptomatic screening or surveillance populations (including supplemental/add-on screening and PHBC surveillance), reported screen-detected cancers with denominators sufficient to calculate cancer detection rate (CDR) or between-modality differences, and used histopathology with or without imaging follow-up as the reference standard (with at least 12 months' follow-up when biopsy was not performed). Diagnostic, recall, problem-solving, symptomatic, and preoperative staging studies were excluded. Only English-language studies were included. Included cohorts reflected contemporary screening ages (generally ≥40 years).

Definition of supplemental and add-on imaging

In this review, supplemental imaging refers to the use of an additional imaging test together with mammography to improve cancer detection. Add-on imaging refers specifically to situations where CEM is performed in addition to mammography or DBT during the same screening visit. This distinction is important because add-on screening may lead to higher recall and biopsy rates since findings from both tests are evaluated.

Conceptual distinction between screening and diagnostic imaging

This review restricted quantitative synthesis to studies evaluating CEM in screening or surveillance populations. Studies conducted in diagnostic, recall, or problem-solving contexts were excluded from meta-analysis because these populations are enriched for malignancy and therefore not representative of population-based screening settings.

Study categorization and screening pathways

To ensure methodologically appropriate synthesis, eligible studies were categorized according to screening pathway and comparator structure:

1. Comparative Screening Studies
   - Direct comparisons of:
     - CEM versus low-energy mammography
     - CEM versus breast MRI
   - Conducted in the same or comparable screening and surveillance populations, including head-to-head or randomized designs
2. Single-arm Screening Studies
   - Reported screening performance of CEM without a parallel comparator
   - Included to contextualize absolute cancer detection but excluded from quantitative pooling due to heterogeneity in screening pathways and populations
3. Add-on Screening Studies
   - Compared mammography-based screening [(digital mammography or DBT] with and without the addition of CEM in the same screening population
   - Incremental cancer detection rate (ICDR) attributable to CEM was assessed

Inclusion of Personal History of Breast Cancer (PHBC) Populations

Women with a PHBC were included because contemporary screening paradigms recognize surveillance imaging as a form of screening, particularly given the elevated interval cancer risk in this population.[10] These cohorts were analyzed within their respective screening pathways and were not intended to redefine screening performance in average-risk populations. PHBC studies were not used to infer screening performance in average-risk populations and were interpreted separately within surveillance and elevated-risk screening contexts.

The characteristics of the included screening and surveillance studies are summarized in Table 1. Additional study-level technical characteristics are provided in Table S1.

Data extraction

Data were independently extracted by two reviewers with previous breast imaging experience using a standardized form. Extracted variables included:

- Study characteristics: author, year, country, study design
- Screening pathway and risk population
- Comparator modality, when applicable
- Unit of analysis (women or examinations)
- Number of screened women or examinations
- Number of screen-detected cancers
- CDR
- ICDR
- Recall rate, biopsy rate, and biopsy yield (PPV3), when reported

Discrepancies were resolved by consensus.

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026

Risk of bias assessment

Risk of bias was assessed using the Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) tool, adapted for screening studies. The following domains were evaluated:

1. Patient selection
2. Index test (CEM)
3. Reference standard
4. Flow and timing

Each domain was rated as low, high, or unclear risk of bias, with applicability concerns assessed separately. Given the nature of screening studies, particular attention was paid to patient selection and completeness of follow-up. Risk-of-bias assessments were summarized qualitatively and tabulated in the supplementary material.

Statistical analysis

Quantitative synthesis was performed using random-effects models to account for between-study variability. The primary effect measure for all comparative analyses was the risk difference (RD) in cancer detection, expressed as additional cancers detected per 1,000 women screened.

Risk difference was selected because:

- CDR is a single-arm outcome dependent on the underlying population risk, and pooling absolute CDRs across heterogeneous screening and surveillance populations would be inappropriate
- RD provides a clinically interpretable estimate of incremental screening benefit
- In add-on screening studies, RD is mathematically equivalent to the ICDR

Statistical heterogeneity was assessed using the I^2 statistic, with values interpreted descriptively. Sensitivity analyses were conducted when the comparator structure or attribution methods differed.

Because of the limited number of studies and variability in design and cancer event counts, quantitative pooling of studies comparing CEM with MRI was not performed. These studies were summarized qualitatively. Single-arm screening studies were summarized descriptively and not pooled due to heterogeneity in populations, screening pathways, and outcome reporting.

All analyses were performed in R (version 4.3.2; R Foundation for Statistical Computing, Vienna, Austria) using the metafor package (version 4.6-0). Statistical significance was defined as a two-sided *P-value* <0.05.

Results

Study selection and characteristics

Fourteen studies met the eligibility criteria and were included in the qualitative synthesis.[20–33] Seven studies contributed data to one or more quantitative meta-analyses.[22,23,27,28,31–33] Included studies evaluated CEM across a range of screening pathways, including primary screening, supplemental screening, and surveillance screening in women with a PHBC. Study populations encompassed average-risk, intermediate-risk, elevated-risk, and post-treatment surveillance cohorts.

Most studies were prospective (12/14), with the remainder being retrospective observational cohorts. Screening contexts included primary screening, supplemental screening, add-on screening to mammography or DBT, and surveillance screening in women with PHBC. Two large prospective add-on screening trials from the United States—the TOCEM study[31] and the SCEMAM study[22]—evaluated the incremental contribution of CEM when added to DBT in PHBC and elevated-risk populations, respectively. Several prospective studies have provided direct comparative data between CEM and low-energy mammography and/or MRI in screening or supplemental screening settings, including cohorts enriched for dense breasts or individuals with elevated risk.[21,23,27–29] The randomized BRAID trial compared CEM with abbreviated MRI in women with dense breasts undergoing supplemental screening.[30] Surveillance screening in PHBC populations was evaluated in retrospective cohorts by Elder et al. (2023) and Gluskin et al. (2020), who used attribution-based methods to assess the contribution of contrast-enhanced imaging.[32,33] Four studies reported single-arm screening performance of CEM without a parallel comparator, including primary screening,[24] supplemental screening in dense or intermediate-risk populations,[20,26] and surveillance screening in PHBC populations.[25] Across studies, the unit of analysis varied between women and examinations, screened sample sizes ranged from several hundred to more than 5,000 participants, and mean or median ages were generally in the mid-50 s to early 60 s. Histopathologic confirmation with imaging follow-up served as the reference standard in nearly all studies, and image interpretation was predominantly performed by breast-imaging–trained or fellowship-trained radiologists.

The study selection process is shown in Figure 1.

Comparative performance: contrast-enhanced mammography (CEM) versus low-energy mammography

Three prospective screening studies directly compared CEM with low-energy mammography in the same screened populations and formed the primary analysis.[23,27,28] These studies included women with dense breasts, as well as average-risk and intermediate-risk cohorts undergoing screening.

All three studies reported higher cancer detection with CEM compared with low-energy mammography (Figure 2). In pooled analysis, CEM was associated with 9.3 additional cancers detected per 1,000 women screened relative to low-energy mammography (RD, +9.3 per 1,000; 95% CI: 4.0–14.6; $p < 0.01$), with no observed heterogeneity ($I^2 = 0\%$). Corresponding screening performance outcomes, including recall and biopsy rates, are summarized in Table 2.

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026

Table 1. Characteristics of Included Screening and Surveillance Studies

Study, Year	Study design	Screening pathway	Risk group/population	Comparator modality	Unit of analysis	*n* screened	Mean/median age
Berg, 2024 (TOCEM)	Prospective	Add-on screening	PHBC	DBT	Woman	1,273	60 (mean)
Berg, 2025 (SCEMAM)	Prospective	Add-on screening	Elevated risk	DBT	Woman	601	59 (mean)
Elder, 2023	Retrospective	Surveillance	PHBC	LE (attribution-based)	Woman	1,191	61 (mean)
Gilbert, 2025 (BRAID)	RCT	Supplemental screening	Dense breasts	Abbreviated MRI	Woman	2,035	58 (mean)
Gluskin, 2020	Retrospective	Surveillance	PHBC	LE (attribution-based)	Woman	971	58 (mean)
Hogan, 2022	Retrospective	Surveillance	PHBC	None	Exam	306	54 (mean)
Horvat, 2025	Prospective	Screening	Average–intermediate risk	LE mammography ± US	Woman	468	56 (mean)
Jochelson, 2017	Prospective	Screening	High risk	MRI	Woman	307	52 (mean)
Lawson, 2023	Prospective	Supplemental screening	Elevated risk	MRI	Woman	468	57 (mean)
Nissan, 2024	Prospective	Screening	Dense breasts	LE mammography	Exam	1,246	56 (mean)
Patel, 2024	Prospective	Supplemental (post-DBT)	Elevated risk ± PHBC	None	Woman	460	57 (mean)
Sorin, 2017	Prospective	Supplemental screening	Dense/intermediate risk	None	Exam	611	55 (mean)
Sorin, 2024	Prospective	Primary screening	Intermediate–high risk	None	Exam	5,424	54 (median)
Sung, 2019	Prospective	Screening	Average risk	LE mammography	Woman	904	55 (mean)

Abbreviations: DBT, digital breast tomosynthesis; LE, low-energy; PHBC, previous history of breast cancer; RCT, randomized controlled trial.

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026

Sensitivity analyses incorporating two surveillance studies in PHBC populations that used attribution-based comparator methods yielded similar estimates, indicating consistency of the observed effect across screening contexts (Figure S1).[32,33]

Comparative performance: contrast-enhanced mammography (CEM) versus MRI

Three studies evaluated CEM in comparison with MRI for screening or supplemental screening.[21,29,30] These included a direct comparative study in a high-risk screening cohort, a randomized trial comparing CEM with abbreviated MRI in women with dense breasts, and a supplemental screening study in an elevated-risk population.

Because of heterogeneity in study design, population characteristics, and the limited number of cancer events, quantitative meta-analysis was not performed. Across these studies, MRI detected numerically more cancers in some cohorts, although differences varied by population and screening context. These findings suggest broadly comparable cancer detection between modalities in selected populations but remain insufficient for definitive comparative conclusions.

The findings from these studies were summarized qualitatively in Table 3.

Add-on screening: mammography-based screening with contrast-enhanced mammography (CEM)

Two large prospective screening studies evaluated the incremental contribution of CEM when added to mammography-based screening.[22,31] The TOCEM trial enrolled women with a PHBC undergoing surveillance screening and compared DBT alone with DBT plus CEM across multiple screening rounds.[31] The SCEMAM trial evaluated elevated-risk women eligible for MRI screening and compared DBT alone with DBT plus CEM at baseline screening.[22] In both studies, the addition of CEM increased cancer detection relative to mammography-based screening alone. In TOCEM, nine additional cancers were detected among 1,273 women screened during the first screening round, with four additional cancers detected in subsequent rounds.[31] In SCEMAM, six additional cancers were detected among 601 women screened with the addition of CEM.[22]

In a pooled random-effects meta-analysis, add-on CEM screening resulted in a statistically significant increase in cancer detection of 7.4 additional cancers per 1,000 women screened (RD, +7.4 per 1,000; 95% CI: 3.6–11.2; $P < 0.001$), with no observed heterogeneity ($I^2 = 0\%$) (Figure 3). Recall and biopsy rates were higher with the addition of CEM, consistent with contrast-enhanced supplemental screening strategies. Incremental screening outcomes for add-on CEM are summarized in Table 2.

Corresponding screening performance outcomes across comparative screening pathways, including recall rates, biopsy rates, and positive predictive value, are summarized in Table 2.

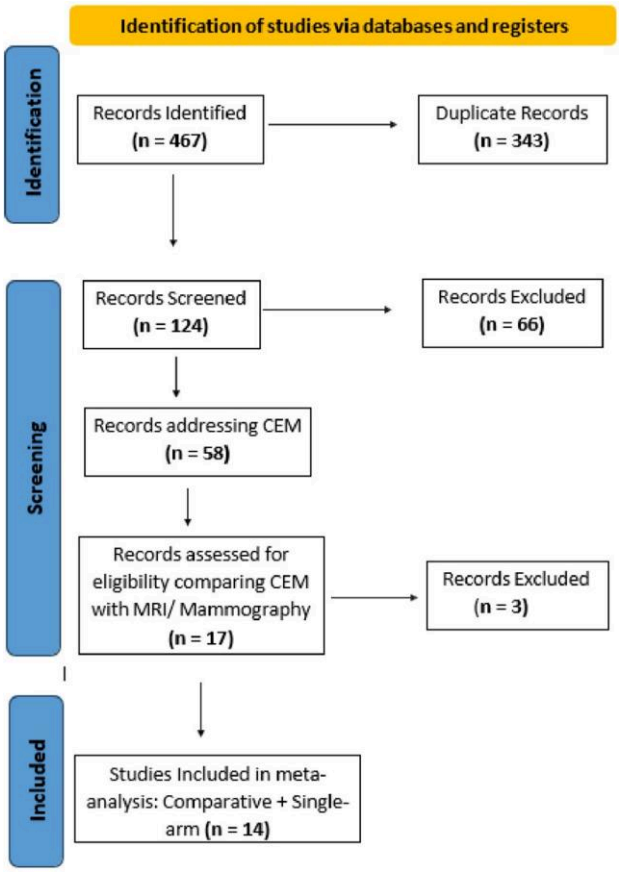


Figure 1. Study selection PRISMA flow diagram. Flow diagram illustrating identification, screening, eligibility assessment, and inclusion of studies in this systematic review and meta-analysis of CEM in screening and surveillance populations. A total of 467 records were identified, of which 124 were screened after duplicate removal. Seventeen studies met eligibility criteria, and 14 studies (comparative and single-arm screening designs) were included in the final quantitative and qualitative synthesis. Abbreviations: CEM, contrast-enhanced mammography.

Single-arm screening performance of contrast-enhanced mammography (CEM)

Four studies reported single-arm screening performance of CEM without a parallel comparator.[20,24–26] These included primary screening, supplemental screening, and surveillance screening populations. Screening outcomes from single-arm studies were summarized descriptively and are presented in Table S2. Reported CDRs ranged from 13.1 to 23.9 cancers per 1,000 examinations, with higher detection observed in elevated-risk and surveillance cohorts. Due to heterogeneity in screening pathways, populations, and outcome reporting, these studies were not pooled quantitatively. Instead, they provide descriptive context regarding the absolute cancer detection achievable with CEM across different screening scenarios.

Recall, biopsy rate and PPV3

In direct screening comparisons with low-energy mammography, CEM was associated with higher recall rates (eg, 18.7% vs 8.3% in dense-breast screening and 18.6% vs

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026

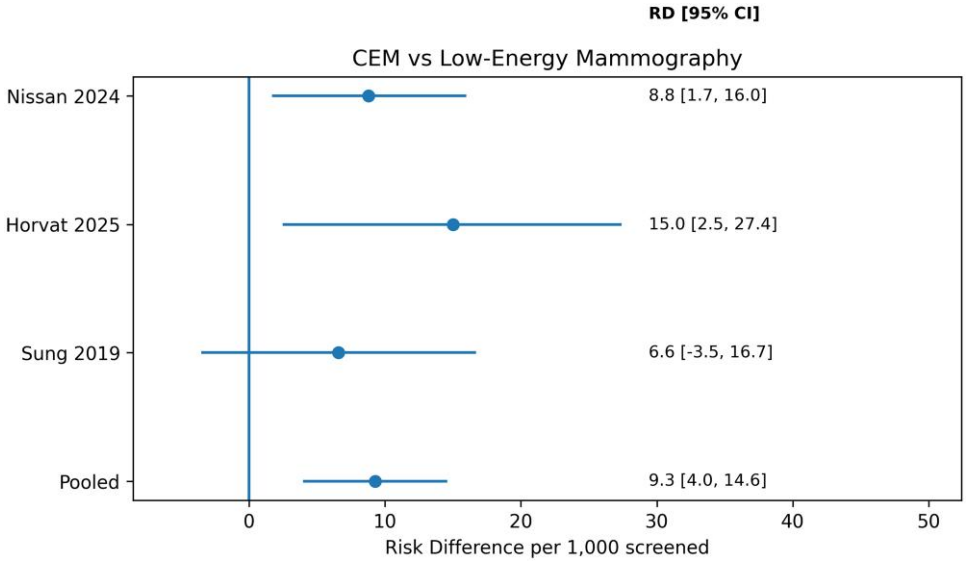


Figure 2. Forest plot of cancer detection comparing contrast-enhanced mammography (CEM) with low-energy mammography. Forest plot showing study-level and pooled risk differences (RD) in cancer detection per 1,000 women screened for contrast-enhanced mammography (CEM) versus low-energy mammography across screening populations. Squares represent point estimates and horizontal lines indicate 95% confidence intervals. The vertical line at zero represents no difference between modalities. The pooled analysis demonstrates increased cancer detection with CEM (RD, +9.3 per 1,000 women; 95% CI, 4.0–14.6) with no observed heterogeneity ($I^2 = 0\%$). Abbreviations: CEM contrast-enhanced mammography; RD, risk difference.

Table 2. Screening Performance Outcomes Across Comparative Screening Pathways

Screening pathway	Outcome	CEM	Comparator	Effect estimate
CEM vs low-energy mammography (primary analysis, screening-only)	Cancer detection rate (/1,000)	12.8–17.1	2.1–4.0	RD +9.3 per 1,000 (95% CI 4.0–14.6)
	Recall rate (%)	18.6–18.7	8.3–10.3	Higher with CEM
	Biopsy rate (%)	4.3–4.8	1.5–2.6	Higher with CEM
	PPV3 (%)	23.9–45.0	14.3–15.6	Higher with CEM
Add-on CEM to mammography-based screening	Incremental cancer detection (/1,000)	4.0–9.0	—	Absolute increase
	Incremental recall (%)	7.8–13.1	—	Absolute increase
	Incremental biopsy (%)	2.7–4.5	—	Absolute increase

Abbreviations: CEM, contrast-enhanced mammography; PPV, positive predictive value.

Table 3. Studies Evaluating **C**ontrast-Enhanced Mammography (CEM) Compared with MRI in Screening Populations

Study	Study Design	Population	Sample Size	Key Findings
Gilbert et al. 2025	Prospective screening study	Women undergoing supplemental imaging following screening recall	~900	MRI demonstrated higher sensitivity; CEM detected cancers not visible on mammography
Jochelson et al. 2017	Prospective pilot screening study	Women at increased risk for breast cancer	307	MRI detected 3 cancers; CEM detected 2 invasive cancers; specificity similar (94.7% vs 94.1%)
Lawson et al. 2023	Prospective screening comparison	Women undergoing screening MRI	~460	MRI showed slightly higher cancer detection; CEM demonstrated comparable specificity

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026

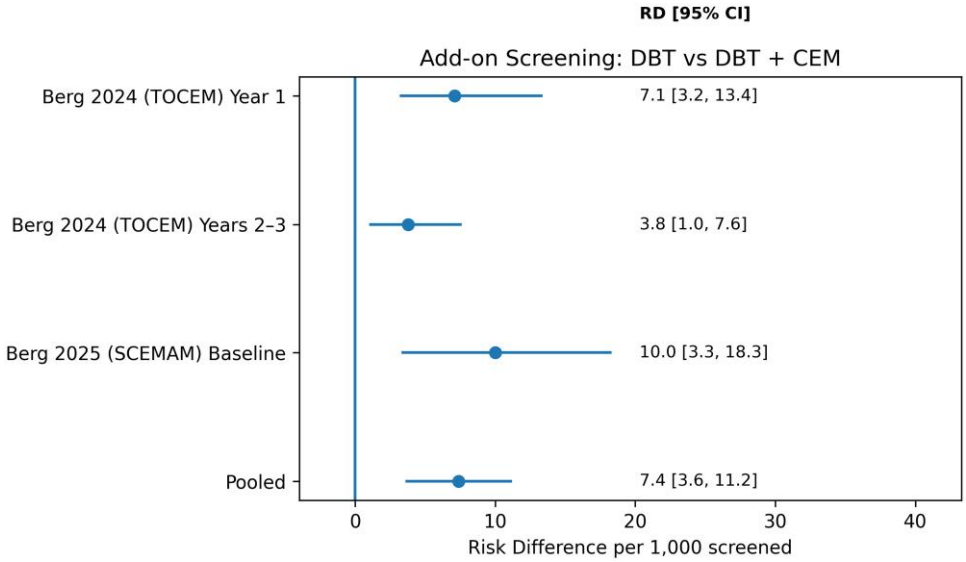


Figure 3. Forest plot of incremental cancer detection with add-on contrast-enhanced mammography (CEM). Forest plot showing study-level and pooled RD in cancer detection per 1,000 women screened for DBT alone versus DBT combined with CEM. Estimates from individual screening rounds of the TOCEM trial and baseline screening in the SCEMAM trial are presented. Squares represent point estimates and horizontal lines indicate 95% confidence intervals. The vertical line at zero represents no incremental detection. The pooled analysis demonstrates increased cancer detection with add-on CEM (RD, +7.4 per 1,000 women; 95% CI, 3.6–11.2) with no observed heterogeneity (I^2 = 0%). Abbreviations: CEM contrast-enhanced mammography; DBT, digital breast tomosynthesis; RD, risk difference.

10.3% in population screening) and higher biopsy rates, with biopsy yield (PPV3) for CEM ranging from 23.9% to 45%, overlapping or exceeding PPV3 observed with low-energy mammography across studies.[23,27,28] In surveillance cohorts, PPV3 for CEM remained moderate to high (eg, 42.9%–47.9% in PHBC screening), despite increased recall.[32,33] In studies that included MRI comparators, recall and biopsy rates varied by study design and screening context. Reported PPV3 values for CEM ranged from 15.4% to 43.8%, while PPV3 for MRI ranged from 14.3% to 35.2% across available studies. Because these comparisons were derived from heterogeneous study designs and populations, direct comparisons of specificity metrics should be interpreted cautiously.[21,29,30] Large prospective add-on screening trials further contextualised these trade-offs: in TOCEM, the addition of CEM to DBT increased recall from 8.1% to 14.7% in the first round and from 3.8% to 8.8% in subsequent rounds, with biopsy rates increasing from 3.5% to 6.8% and 1.7% to 4.4%, respectively, while participant-level PPV3 for DBT plus CEM was 33.3% in year 1 and 17.4% across later rounds.[31] Similarly, in SCEMAM, the addition of CEM increased the false-positive interpretations from 8.1% to 21.6%, with 8.3% of women undergoing biopsy and cancer confirmed in 12% of biopsied women, alongside the detection of six additional cancers (ICDR, 10.0 per 1,000 women) and no interval cancers at 1-year follow-up.[22]

Risk of Bias

Risk of bias was assessed using QUADAS-2 and was judged to be low across most domains for studies included in the primary meta-analyses. Two surveillance studies were included in sensitivity analyses demonstrated unclear risk in the flow and timing domain due to attribution-based comparator assessment.[32,33] Risk of bias across included studies is summarized in Table 4. Detailed QUADAS-2 signalling questions and study-level assessments are shown in Figure S2.

Discussion

To our knowledge, this represents the first systematic review and meta-analysis to explicitly restrict quantitative synthesis to screening cohorts and clearly distinguish screening populations from diagnostic and problem-solving cohorts. This distinction is important because diagnostic cohorts are inherently enriched for malignancy since imaging is performed to evaluate a suspected abnormality. The resulting higher disease prevalence can inflate estimates of test performance and limit the generalizability of these findings to screening practice. In contrast, screening populations involve asymptomatic individuals with substantially lower disease prevalence, where modest absolute increases in cancer detection may still be clinically meaningful but often occur alongside higher recall and biopsy rates. Failure to distinguish between these contexts may therefore overestimate the expected benefit of a screening modality. Similar challenges were observed in early studies of MRI screening, where results from diagnostic or enriched populations initially suggested higher performance than later observed in population-based screening cohorts.[34,35]

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026

Table 4. Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) Risk-of-bias Summary

Study	Patient selection	Index test (CEM)	Reference standard	Flow & timing	Overall risk
Berg 2024 (TOCEM)	Low	Low	Low	Low	Low
Berg 2025 (SCEMAM)	Low	Low	Low	Low	Low
Nissan 2024	Low	Low	Low	Low	Low
Horvat 2025	Low	Low	Low	Low	Low
Sung 2019	Low	Low	Low	Low	Low
Jochelson 2017	Low	Low	Low	Low	Low
Gilbert 2025 (BRAID)	Low	Low	Low	Low	Low
Lawson 2023	Low	Low	Low	Low	Low
Gluskin 2020[†]	Low	Low	Low	Unclear	Moderate
Elder 2023[†]	Low	Low	Low	Unclear	Moderate

Abbreviation: CEM, contrast-enhanced mammography.
[†]Surveillance screening studies in women with a personal history of breast cancer.
Unclear risk for flow and timing reflects variation in follow-up intervals and attribution-based comparator methods rather than concerns regarding study conduct.

The design of screening pathways can influence how performance metrics are interpreted. In add-on screening, CEM is performed in addition to mammography or DBT, and findings from either test can lead to recall or biopsy. Because of this, add-on screening strategies naturally result in higher recall and biopsy rates compared with mammography alone, even when cancer detection improves. Similar patterns have been observed when other supplemental screening methods, such as breast MRI or DBT, were first introduced. Therefore, differences in recall rates should be interpreted in the context of the screening pathway rather than attributed only to the imaging modality itself.[36,37]

Clinical positioning

This pathway-stratified analysis found that CEM increased cancer detection compared with low-energy mammography in screening and surveillance contexts. A small number of comparative studies also suggest that cancer detection with CEM may approach that observed with MRI in selected populations; however, the limited number of studies and heterogeneous designs preclude definitive comparative conclusions. These findings support a context-specific role for CEM within contemporary breast cancer screening pathways, particularly as a supplemental modality to mammography-based screening or as an alternative contrast-based option when MRI is unavailable or not tolerated. Emerging evidence, including studies in women with extremely dense breasts (eg, Nissan et al.), suggests that CEM may provide meaningful additional cancer detection in selected subgroups.[28] However, current evidence remains insufficient to support CEM as a replacement for mammography in average-risk populations or as a definitive non-inferior alternative to MRI in high-risk screening cohorts. Findings from cohorts of women with a PHBC should therefore be interpreted primarily in the context of surveillance screening rather than general population screening.

Recall and biopsy trade-offs

Across screening pathways, increased cancer detection with CEM was consistently accompanied by higher recall and biopsy rates, an expected trade-off of contrast-enhanced supplemental imaging. In direct comparisons with low-energy mammography, CEM showed higher recall and biopsy rates while maintaining biopsy yield within acceptable ranges, with PPV3 values generally overlapping or exceeding those of mammography. In surveillance and elevated-risk cohorts, PPV3 for CEM remained moderate to high despite increased recall. In studies that included MRI comparators, recall and biopsy rates varied by study and screening context. Reported biopsy yield (PPV3) for CEM fell within ranges similar to those reported for MRI across individual studies; however, differences in study design, screening pathways, and population risk profiles limit direct comparison between modalities. Large prospective add-on trials further contextualized these findings: the addition of CEM to DBT increased recall and biopsy rates but preserved acceptable biopsy yield and enabled detection of additional cancers without an observed increase in interval cancers at short-term follow-up. Collectively, these data support a balanced interpretation of CEM screening trade-offs, consistent with other supplemental screening modalities.

The QUADAS-2 assessment indicates that the available screening and surveillance CEM literature is methodologically robust with respect to image interpretation and outcome verification. Importantly, studies of women with a PHBC were treated as surveillance screening cohorts, consistent with contemporary guideline frameworks and distinct from diagnostic or problem-solving populations.

Comparison with prior evidence

Prior systematic reviews of CEM have frequently pooled heterogeneous populations, including diagnostic, screening, and preoperative settings, limiting applicability to population-

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026

based screening practice.[15] By restricting quantitative synthesis to true screening cohorts and stratifying analyses by screening pathway, this study addresses a key methodological limitation of earlier work and provides more interpretable estimates for clinical and policy use. While some prior studies have shown MRI detecting numerically more cancers in specific cohorts, the small number of available comparative screening studies and differences in study design and populations in the present study limit definitive conclusions regarding relative performance. These findings highlight the importance of interpreting modality comparisons within the context of specific screening populations and pathways.[23,28]

Compared with other emerging screening technologies, CEM occupies a distinct position. DBT primarily improves lesion conspicuity through structural, tomographic separation of overlapping tissue, thereby reducing the masking effect of breast density,[38] whereas CEM improves lesion conspicuity through functional contrast enhancement related to tumor neoangiogenesis.[39] MRI remains the most sensitive contrast-based screening modality and has the strongest evidence base in high-risk populations. However, CEM may offer a more accessible contrast-enhanced option in selected settings, particularly when MRI is unavailable, contraindicated, or poorly tolerated. In this way, CEM may fit most naturally into risk-stratified screening pathways as a supplemental option for women with dense breasts, intermediate risk, or selected surveillance populations, rather than as a universal replacement for either mammography or MRI.[40,41] Practical implementation also requires consideration of trade-offs: compared with standard mammography, CEM involves a modest increase in radiation dose[42] and a low but non-zero risk of iodinated contrast reactions,[43] while potentially offering lower cost and greater availability than MRI.[35] These factors are likely to be especially relevant in health systems where MRI capacity is limited and contrast-based screening access is uneven.

Strengths and limitations

The strengths of this study include restriction to screening and surveillance populations, *a priori* stratification by clinically relevant screening pathways, use of absolute RD to enhance clinical interpretability, and formal assessment of risk of bias using standardized methods. These design choices improve the relevance of findings for clinical practice and guideline development.

Limitations include the limited number of direct CEM versus MRI comparisons and the heterogeneity of screened populations, including elevated-risk and PHBC subgroups. Because surveillance cohorts of women with a PHBC have a higher baseline risk than average-risk screening populations, including these studies may influence absolute cancer detection estimates; however, pathway-stratified and sensitivity analyses incorporating attribution-based surveillance comparisons yielded consistent effect estimates across screening contexts. Restriction to English-language publications may have introduced selection bias. Variability in reader experience and institutional protocols may influence observed performance metrics, and long-term clinical outcomes remain uncertain.

Another limitation across studies was heterogeneity in follow-up duration and interval cancer ascertainment, which may result in optimistic sensitivity estimates. This limitation underscores the need for ongoing prospective trials and multiround screening data to more precisely define interval cancer rates and long-term outcomes associated with CEM-based screening pathways. Moreover, this analysis focuses on screen-detected cancers and does not address long-term outcomes, such as interval cancer rates, stage shifts, or mortality reductions.

Conclusions

When evaluated exclusively in screening contexts, CEM provides meaningful incremental cancer detection as an add-on to mammography-based screening. Available comparative studies suggest that cancer detection with CEM may be broadly similar to that observed with MRI in selected screening populations; however, the limited number of studies and heterogeneity in design preclude definitive comparative conclusions. These findings support a targeted role for CEM within contemporary screening pathways—particularly where MRI access is limited—while underscoring the need for further research addressing long-term outcomes, optimal implementation, and patient-centred trade-offs. By clarifying where CEM adds value and where uncertainty persists, this work aims to inform balanced, evidence-based screening decisions rather than advocate for universal adoption.

Funding

None declared.

Conflict of interest statement

None declared.

Author contributions

All authors contributed to the conception, design, analysis, drafting, and final approval of the manuscript. K.V.S. contributed the maximum among all. Komal Saluja (Conceptualization, Data curation, Project administration, Writing—original draft), Sangeeta Saxena (Conceptualization, Writing—review & editing), Harshvardhan Khokhar (Visualization, Writing—review & editing), Ashok Sharma (Conceptualization, Writing—review & editing), Dharmraj Meena (Conceptualization, Writing—review & editing), Atul Sharma (Conceptualization, Writing—review & editing), Mayank Goyal (Conceptualization, Supervision, Writing—review & editing)

Declaration of AI use

During the preparation of this work, the authors used ChatGPT and Grammarly to improve the clarity and

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026

readability of the text. The output of the AI tool was reviewed and edited by the authors, who take full responsibility for the content of the publication.

Data availability

All data analyzed in this study were obtained from previously published articles and are included within the manuscript and Supplementary Material.

Supplementary material

Supplementary material is available at *Journal of Breast Imaging* online.

References

1. Bray F, Laversanne M, Sung H, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin*. 2024;74(3):229-263. doi:10.3322/caac.21834
2. American Cancer Society. Key Statistics for Breast Cancer. Accessed January 11, 2026. https://www.cancer.org/cancer/types/breast-cancer/about/how-common-is-breast-cancer.html
3. Salim M, Dembrower K, Eklund M, Lindholm P, Strand F. Range of Radiologist Performance in a Population-based Screening Cohort of 1 Million Digital Mammography Examinations. *Radiology*. 2020;297(1):33-39. doi:10.1148/radiol.2020192212
4. Boyd NF, Guo H, Martin LJ, et al. Mammographic Density and the Risk and Detection of Breast Cancer. *N Engl J Med*. 2007;356(3):227-236. doi:10.1056/NEJMoa062790
5. Kerlikowske K, Zhu W, Tosteson ANA, et al. Identifying Women With Dense Breasts at High Risk for Interval Cancer. *Ann Intern Med*. 2015;162(10):673-681. doi:10.7326/M14-1465
6. Important Information: Final Rule to Amend the Mammography Quality Standards Act (MQSA). U.S. Food and Drug Administration. Updated September 10, 2024. Accessed January 11, 2026. https://www.fda.gov/radiation-emitting-products/mammography-quality-standards-act-mqsa-and-mqsa-program/important-information-final-rule-amend-mammography-quality-standards-act-mqsa
7. Oeffinger KC, Fontham ETH, Etzioni R, et al. Breast Cancer Screening for Women at Average Risk: 2015 Guideline Update From the American Cancer Society. *JAMA*. 2015;314(15):1599. doi:10.1001/jama.2015.12783
8. Nicholson WK, Silverstein M, Wong JB, et al. Screening for Breast Cancer: US Preventive Services Task Force Recommendation Statement. *JAMA*. 2024;331(22):1918-1930. doi:10.1001/JAMA.2024.5534
9. Monticciolo DL, Newell MS, Moy L, Niell B, Monsees B, Sickles EA. Breast Cancer Screening in Women at Higher-Than-Average Risk: Recommendations From the ACR. *J Am Coll Radiol*. 2018;15(3):408-414. doi:10.1016/j.jacr.2017.11.034
10. Mainiero MB, Moy L, Baron P, et al. ACR Appropriateness Criteria® Breast Cancer Screening. *J Am Coll Radiol*. 2017;14(11S):S383-S390. doi:10.1016/j.jacr.2017.08.044
11. Ding W, Fan Z, Xu Y, et al. Magnetic resonance imaging in screening women at high risk of breast cancer: A meta-analysis. *Medicine (Baltimore)*. 2023;102(10):e33146. doi:10.1097/MD.0000000000033146
12. Fields BKK, Joe BN. Screening Breast MRI Effectively Detects Early-Stage Breast Cancer in High-Risk Patients without Prior History of Breast Cancer. *Radiol Imaging Cancer*. 2024;6(2):e249005. doi:10.1148/rycan.249005
13. Heller SL, Moy L. Breast MRI Screening: Benefits and Limitations. *Curr Breast Cancer Rep*. 2016;8(4):248-257. doi:10.1007/s12609-016-0230-7
14. Jochelson MS, Lobbes MBI. Contrast-enhanced Mammography: State of the Art. *Radiology*. 2021;299(1):36-48. doi:10.1148/radiol.2021201948
15. Zhu X, Huang J ming Zhang K, et al. Diagnostic Value of Contrast-Enhanced Spectral Mammography for Screening Breast Cancer: Systematic Review and Meta-analysis. *Clin Breast Cancer*. 2018;18(5):e985-e995. doi:10.1016/j.clbc.2018.06.003
16. Gennaro G, Cozzi A, Schiaffino S, Sardanelli F, Caumo F. Radiation Dose of Contrast-Enhanced Mammography: A Two-Center Prospective Comparison. *Cancers (Basel)*. 2022;14(7):1774. doi:10.3390/cancers14071774
17. Wright K, Shkabari B, Booth C, Knopf K, Tregear M, Gyawali B. Congruence of cancer screening recommendations between the USPSTF and the top ten US cancer centers: a cross sectional study. *EClinicalMedicine*. 2025;82:103169. doi:10.1016/j.eclinm.2025.103169
18. Imai M, Nakamura Y, Yoshino T. Transforming cancer screening: the potential of multi-cancer early detection (MCED) technologies. *Int J Clin Oncol*. 2025;30(2):180-193. doi:10.1007/s10147-025-02694-5
19. Salameh JP, Bossuyt PM, McGrath TA, et al. Preferred reporting items for systematic review and meta-analysis of diagnostic test accuracy studies (PRISMA-DTA): explanation, elaboration, and checklist. *BMJ*. 2020:370:m2632. doi:10.1136/bmj.m2632
20. Patel BK, Carnahan MB, Northfelt D, et al. Prospective Study of Supplemental Screening With Contrast-Enhanced Mammography in Women With Elevated Risk of Breast Cancer: Results of the Prevalence Round. *J Clin Oncol*. 2024;42(32):3826-3836. doi:10.1200/JCO.22.02819
21. Jochelson MS, Pinker K, Dershaw DD, et al. Comparison of screening CEDM and MRI for women at increased risk for breast cancer: A pilot study. *Eur J Radiol*. 2017;97:37-43. doi:10.1016/j.ejrad.2017.10.001
22. Berg WA, Vargo A, Lu AH, et al. Screening for Breast Cancer with Contrast-enhanced Mammography as an Alternative to MRI: SCEMAM Trial Results. *Radiology*. 2025;315(3):e242634. doi:10.1148/radiol.242634
23. Horvat JV, Amir T, Watt GP, et al. Comparison of Contrast-enhanced Mammography and Low-Energy Imaging with or without Supplemental Whole-Breast US in Breast Cancer Detection. *Radiology*. 2025;314(3):e242006. doi:10.1148/radiol.242006
24. Sorin V, Rahman N, Halabi N, Barash Y, Klang E, Sklair-Levy M. Evaluating ten years of breast cancer screening with contrast enhanced mammography in women with Intermediate-high risk. *Eur J Radiol*. 2024;181:111807. doi:10.1016/j.ejrad.2024.111807
25. Hogan MP, Amir T, Sevilimedu V, Sung J, Morris EA, Jochelson MS. Contrast-enhanced digital mammography screening for intermediate-risk women with a history of lobular neoplasia.

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026

AJR Am J Roentgenol. 2021;216(6):1486-1491. doi:10.2214/AJR.20.23480
26. Sorin V, Yagil Y, Yosepovich A, et al. Contrast-enhanced spectral mammography in women with intermediate breast cancer risk and dense breasts. *AJR Am J Roentgenol*. 2018;211(5):W267-W274. doi:10.2214/AJR.17.19355
27. Sung JS, Lebron L, Keating D, et al. Performance of dual-energy contrast-enhanced digital mammography for screening women at increased risk of breast cancer. *Radiology*. 2019;293(1):81-88. doi:10.1148/radiol.2019182660
28. Nissan N, Comstock CE, Sevilimedu V, et al. Diagnostic Accuracy of Screening Contrast-enhanced Mammography for Women with Extremely Dense Breasts at Increased Risk of Breast Cancer. *Radiology*. 2024;313(1):e232580. doi:10.1148/radiol.232580
29. Lawson MB, Partridge SC, Hippe DS, et al. Comparative Performance of Contrast-enhanced Mammography, Abbreviated Breast MRI, and Standard Breast MRI for Breast Cancer Screening. *Radiology*. 2023;308(2):e230576. doi:10.1148/radiol.230576
30. Gilbert FJ, Payne NR, Allajbeu I, et al. Comparison of supplemental breast cancer imaging techniques—interim results from the BRAID randomised controlled trial. *Lancet*. 2025;405(10493):1935-1944. doi:10.1016/S0140-6736(25)00582-3
31. Berg WA, Berg JM, Bandos AI, et al. Addition of Contrast-enhanced Mammography to Tomosynthesis for Breast Cancer Detection in Women with a Personal History of Breast Cancer: Prospective TOCEM Trial Interim Analysis. *Radiology*. 2024;311(1):e231991. doi:10.1148/radiol.231991
32. Elder K, Matheson J, Nickson C, et al. Contrast enhanced mammography in breast cancer surveillance. *Breast Cancer Res Treat*. 2023;199(2):221-230. doi:10.1007/s10549-023-06916-0
33. Gluskin J, Saccarelli CR, Avendano D, et al. Contrast-enhanced mammography for screening women after breast conserving surgery. *Cancers (Basel)*. 2020;12(12):3495. doi:10.3390/cancers12123495
34. Murad MH, Lin L, Chu H, et al. The association of sensitivity and specificity with disease prevalence: analysis of 6909 studies of diagnostic test accuracy. *CMAJ*. 2023;195(27):E925-E931. doi:10.1503/cmaj.221802
35. Coffey K, Jochelson MS. Contrast-enhanced mammography in breast cancer screening. *Eur J Radiol*. 2022;156:110513. doi:10.1016/j.ejrad.2022.110513
36. Ontario Health (Quality). Supplemental Screening as an Adjunct to Mammography for Breast Cancer Screening in People With Dense Breasts: A Health Technology Assessment. *Ont Health Technol Assess Ser*. 2023;23(9):1-293.
37. Kuhl CK, Schrading S, Strobel K, et al. Supplemental breast MR imaging screening of women with average risk of breast cancer. *Radiology*. 2017;283(2):361-370. doi:10.1148/radiol.2016161444
38. Kopans DB. Digital breast tomosynthesis from concept to clinical care. *AJR Am J Roentgenol*. 2014;202(2):299-308. doi:10.2214/AJR.13.11520
39. Polat DS, Evans WP, Dogan BE. Contrast-enhanced digital mammography: technique, clinical applications, and pitfalls. *AJR Am J Roentgenol*. 2020;215(5):1267-1278. doi:10.2214/AJR.19.22412
40. Phillips J, Mihai G, Hassonjee SE, et al. Comparative dose of contrast-enhanced spectral mammography (CESM), digital mammography, and digital breast tomosynthesis. *AJR Am J Roentgenol*. 2018;211(4):839-845. doi:10.2214/AJR.17.19036
41. van Nijnatten TJA, Schipper RJ, Lobbes MBI. Barriers to implementation of contrast-enhanced mammography in clinical practice. *AJR Am J Roentgenol*. 2023;220(2):289-296. doi:10.2214/AJR.22.28567
42. Berg WA, Rafferty EA, Friedewald SM, Hruska CB, Rahbar H. Screening algorithms in dense breasts: AJR expert panel narrative review. *AJR Am J Roentgenol*. 2020;216(2):275-294. doi:10.2214/AJR.20.24436
43. Marcon M, Fuchsjäger MH, Clauser P, Mann RM. ESR Essentials: screening for breast cancer - general recommendations by EUSOBI. *Eur Radiol*. 2024;34(10):6348-6357. doi:10.1007/s00330-024-10740-5

Downloaded from academic.oup.com/jbi/article/8/4/354/8739296 by Universidad El Bosque user on 24 September 2026